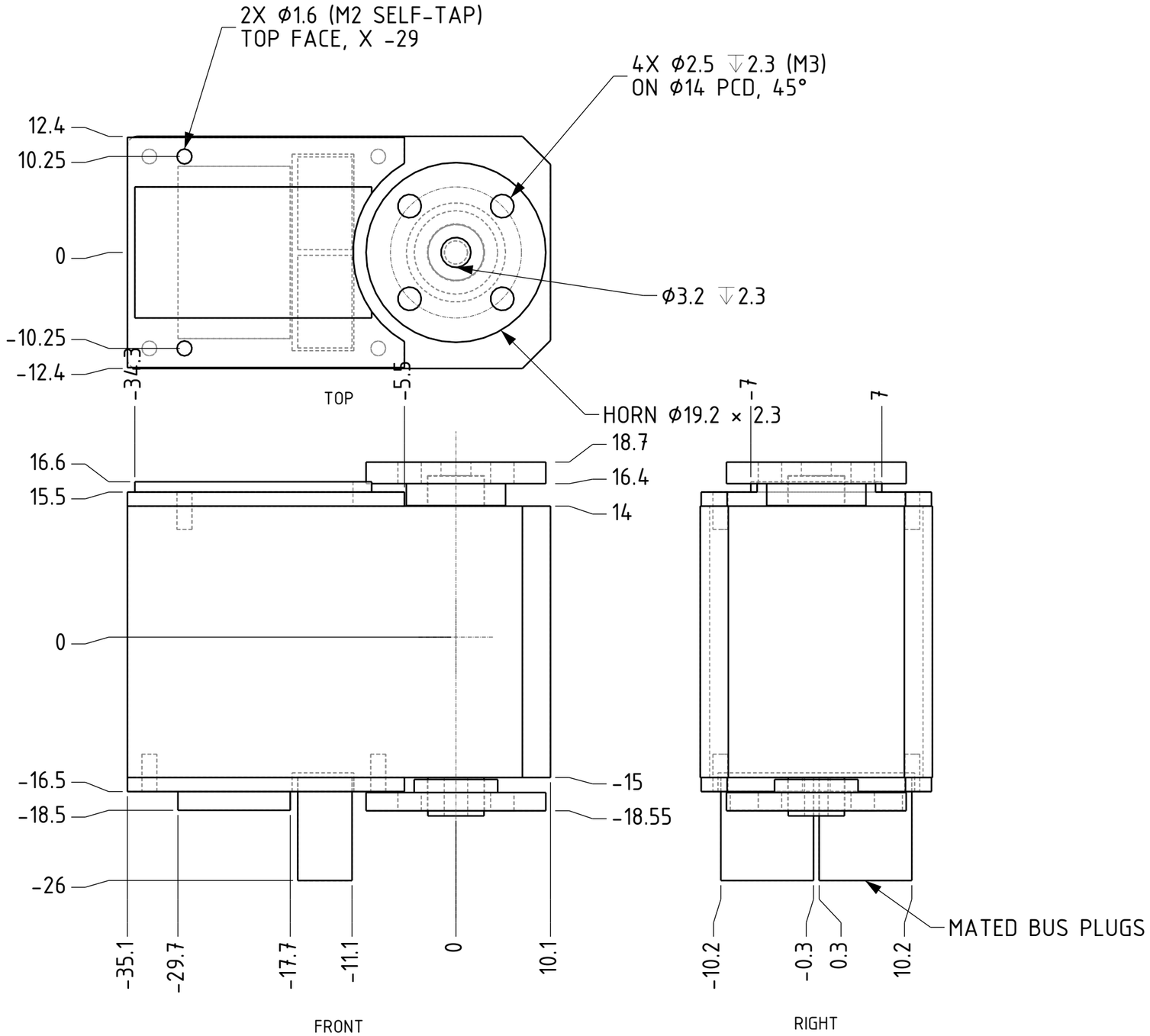


SERVO (servo frame)

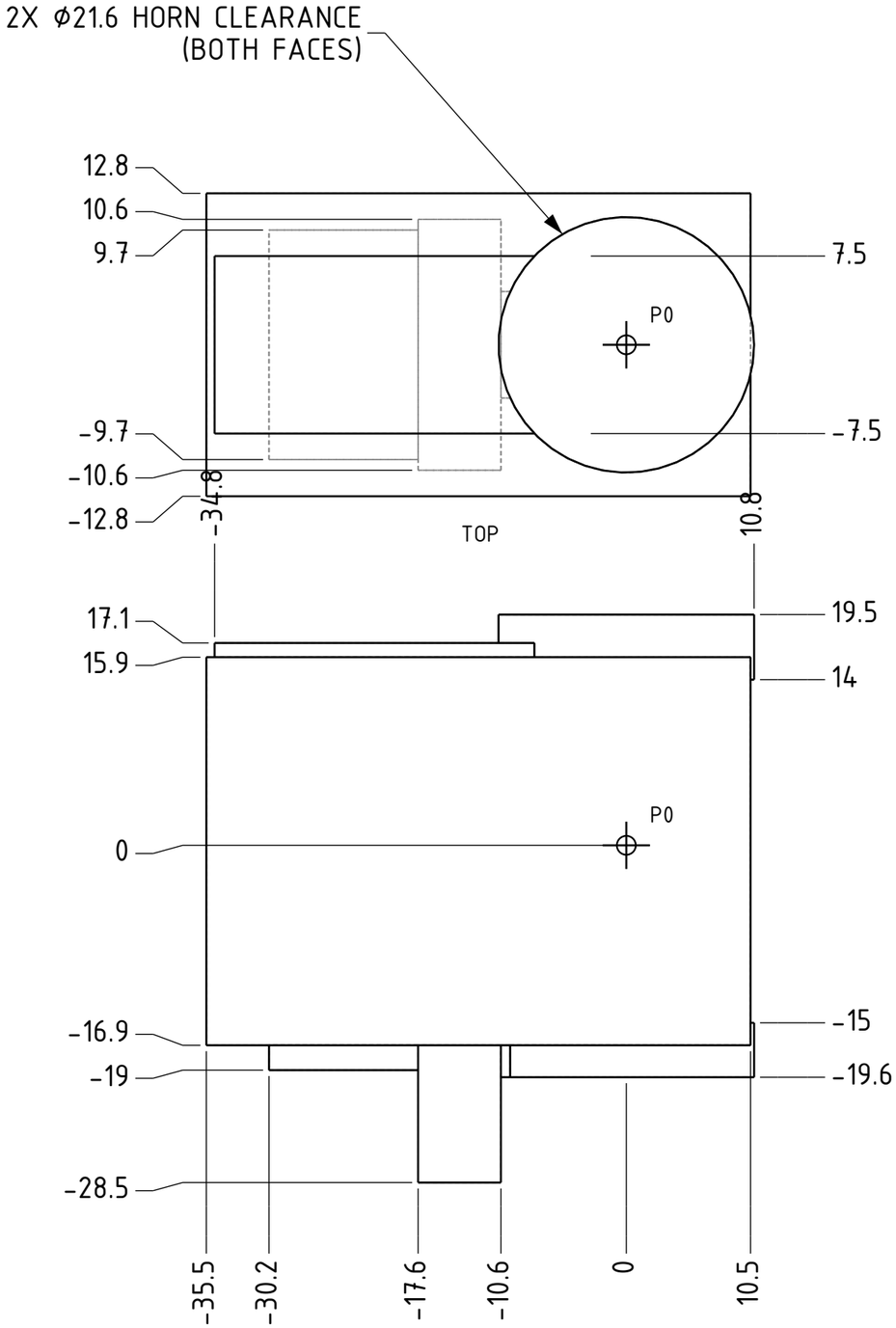
1. Case 45.2 × 24.8, x −35.1 to 10.1, full outline z −15 to 14; +X end corners C3, −X end corners R1.
2. Top face (output side) z 15.5, from x −35.1 to −5.5, stopping at R11 round the shaft. Rib on it: x −34.3 to −9, y ±7, up to z 16.6, its shaft end at R11.
3. Bottom face (connector side) z −16.5, from x −35.1 to −5.5, stopping at R11. Bump on it: x −29.7 to −17.7, y ±9.2, down to z −18.5.
4. Connector sockets x −16.5 to −11.5, y 0.5 to 10 and −10 to −0.5; the plugs go in from the bottom face. Mated plugs modelled to z −26.
5. Output horn ⌀19.2: disc z 16.4 to 18.7, hub ⌀10.6 down to the case (z 14); 4X ⌀2.5 M3 on ⌀14 PCD at 45°, centre ⌀3.2. Bare spline (horn off) ⌀6 up to z 17.2.
6. Passive horn ⌀19.2: disc z −16.6 to −18.55, turning on the idler boss ⌀6 (tip z −19.1).
7. Case holes ⌀1.6 (M2 self-tap): top face at (−29, ±10.25), bottom face at (−32.75, ±10.25). Each face has a second pair at x −8.3; the base uses the bottom one.

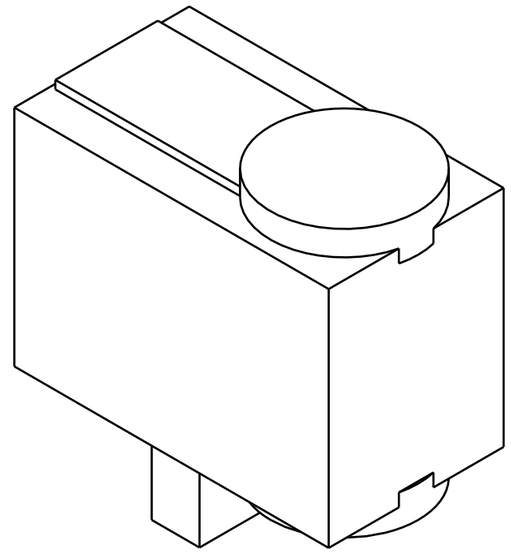
NOTES

- Bought part: Feetech STS3215 (the 7.4 V and 12 V variants share this case). Sizes measured from Feetech's CAD inside SO-101's assembly STEP (cad/reference/so101/step/STS3215_from_SO101.step); each modelled size is the real one or up to 0.25 bigger. servo_reference.py checks every pocket against that model.
- Both horns come off (one centre screw each): every servo goes into its part with its horns off. Sheet 2 gives pocket P, the envelope every servo part cuts.

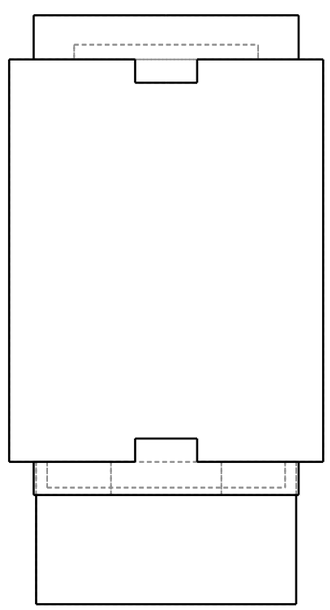
THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART STS3215 servo (reference) (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM 6	DATE 2026-11-10
MATERIAL bought part	SHEET 2 of 21





ISOMETRIC (NOT TO SCALE)



RIGHT

POCKET P (servo frame; 0.4 / 0.5 clearance included)

1. Case box x −35.5 to 10.5, y ±12.8, z −16.9 to 15.9: the case + 0.4 all round (printing tolerance).
2. Rib channel x −34.8 to −3.5 (into the spline channel), y ±7.5, z 15.5 to 17.1; bump channel x −30.2 to −17.2, y ±9.7, z −19 to −16.5 (each + 0.5).
3. Plug bay box x −17.6 to −10.6, y ±10.6, z −28.5 to −15 (mated plugs + cable bend).
4. Relief box x −10.6 to −9.0, y ±4.5, z −19.6 to −15: no knife edge where the plug bay meets the idler ⌀21.6.
5. Idler cylinder ⌀21.6 on the shaft axis, z −19.6 to −15 (passive horn and boss); horn cylinder ⌀21.6, z 14 to 19.5 (output horn; bare spline ⌀7 to z 17.6).

   Way in (each part sheet): the case box, rib, bump and plug bay boxes run on out along the servo's entry direction, and the spline and boss ⌀7 become slots of their width. The servo goes in with both horns off.

NOTES

- Each part sheet locates P0 and gives the servo axes in the part frame; cut the union of the bodies.
- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B

PROJECT TALOS 5-axis arm	EST. MASS -
PART Servo pocket P (cut into parts) (part sheet 2 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-10
MATERIAL -	SHEET 3 of 21